

Mind the Cap: Output-Budget Regimes Change the Measured Multilingual Reasoning Gap

Anonymous ACL submission

Abstract

Multilingual evaluations report accuracy at a single output-token cap, but languages need different numbers of tokens to express the same content, which makes that cap a hidden experimental variable. We test whether the native-vs-translate gap on MGSM (German, Thai, Swahili) is a token-budget artifact, sweeping the budget rather than fixing it, for Qwen3-8B and Llama-3.1-8B-Instruct under four prompting strategies. The measured gap moves by up to 38.9 points across budgets, and at tight caps length normalization can reverse which strategy scores higher. We pre-registered the sweep’s three Qwen peaks and its near-zero value at 1024, then tested them on 540,000 independently hard-capped decodes; a second frozen family of six Holm-corrected tests rejects every null. The prospectively frozen test at $B^* = 1024$ still fails to reject, because native accuracy has already saturated there; above saturation the residual difference is a strategy-performance gap, not an identified reasoning deficit. The same truncation channel prices a cost-ordered adaptation ladder: a cross-fitted Thai vocabulary extension closes 0.00 points of the gap at the frozen budget and 4.9 points where 19% of traces still truncate. Treat the output cap as an independent variable and report accuracy across the budget regime, not at a single budget.

1 Introduction

A recurring finding in multilingual LLM evaluation is that models solve reasoning problems better when the problem is first translated into English than when they reason in the original language. This “multilingual reasoning gap” is usually measured at one generation length. Languages differ in how many tokens they need to express parallel content, and strategies differ in when they emit their answer. Under a hard output cap, both features interact with the cap in ways that can resemble a strategy-performance difference.

We ask a narrow, falsifiable question: is the native-vs-translate gap on MGSM a token-budget artifact? We operationalize a budget artifact as the change in that gap between a token-matched cap and a length-normalized, FLORES-200-premium-adjusted cap. If giving NATIVE its premium-scaled budget closes the gap, token matching contributed to the measured difference; if the gap persists, this particular correction does not explain it.

Our contribution is methodological:

1. The measured contrast can depend strongly on the evaluation budget. We pair a prospectively frozen non-rejection at one budget with a retrospective sweep that locates the budget-binding regime. We then replicate that budget dependence under a pre-registered held-out design in which every budget is decoded under its own hard cap.
2. Under tight caps, length normalization can reverse which strategy looks better.
3. Five ledger-computable audits—trace-language ID, COMET translation quality, parser robustness, decoder parity, and normalizer sensitivity—bound the interpretations the stored ledger supports.
4. The same truncation channel prices a cost-ordered adaptation ladder: raising the serving budget and adding language-specific tokens act on the same traces, and a directly measured, cross-fitted vocabulary extension closes the gap only where the cap still truncates.

Prompt language, reformulation, answer-format compliance, translation quality, and reasoning-trace language are confounded in this design. The object of inference is therefore strategy performance under controlled prefix budgets, not causal reasoning ability.

2 Related work

This study connects multilingual chain-of-thought evaluation on MGSM (Shi et al., 2023) with work on test-time compute, output budgeting, and budget-forcing, including s1 (Muennighoff et al., 2025). Our length proxy follows the FLORES parallel-text lineage and NLLB’s FLORES-200 benchmark (Goyal et al., 2022; NLLB Team et al., 2022), while the measurement checks use GlotLID (Kargaran et al., 2023) and COMET (Rei et al., 2022). Cross-lingual tokenization disparities make output length a language-specific computational cost (Ahia et al., 2023; Petrov et al., 2023). We study how a prefix budget changes a multilingual strategy contrast after language-length normalization. A common response is to extend the vocabulary with target-language pieces and continue pre-training (Cui et al., 2023; Fujii et al., 2024). Principled initialization of the new embeddings is one refinement (Dobler and de Melo, 2023), and §6 prices that intervention against our estimand.

Inference-budget work likewise finds that truncating reasoning harms accuracy in model- and modality-dependent ways (Su et al., 2026), while training models to recover answers from truncated traces could shrink the budget-binding regime (Silvestri and Cetin, 2026). Multilingual studies find that English-pivot reasoning can improve accuracy yet drift off target (Luo et al., 2025), and identify input comprehension as a major loss source addressed by an English-intermediary pipeline (Ko et al., 2025). Training-side methods further shrink native-pivot gaps to roughly 2–3.5% under matched supervision (Lasbordes et al., 2026) and produce cross-lingual self-distillation gains that grow with inference budget (Liu et al., 2026). Our contribution is orthogonal and evaluational: these methods should be measured under budget sweeps that reveal whether supervision, strategy, or answer-emission timing drives the reported gap.

3 Design and estimands

Data and strategies. MGSM has 250 items per language in German (de), Thai (th), and Swahili (sw), three languages that span a wide range of FLORES-200 token premiums. We evaluate four strategies, also called arms. NATIVE reasons and answers in language L . TRANSLATE-ACT translates the problem to English, then solves it. PIVOT reasons in English and answers in L . CODE-SWITCHED uses an English scaffold. We draw $k = 8$ samples

per item from Qwen3-8B (Qwen Team, 2025) (confirmatory) and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) (secondary, not a replication), for 48,000 stored generations.

Prefix-defined budgets. Each (item, sample) has one stored 4096-token generation. We call this store of generations, together with the scores of all their prefixes, the *ledger*. Evaluating budget B means scoring the length- B prefix. This removes between-budget sampling noise but limits conclusions to prefix-defined evaluation. The retrospective regime sweep uses $B \in \{64, 128, 192, 256, 384, 512, 768, 1024\}$; 2048 and 4096 are added only for extended crossover and best-arm checks.

Independent decoding. Because prefix budgets share a trajectory, we regenerated the sweep with decodes that were hard-capped one budget at a time. The stored ledger becomes a discovery sample: its published sweep values were pre-registered as point predictions in prereg-independent-decoding.md, frozen at git tag independent-protocol-freeze before any of the new generation ran, and then tested on a fresh confirmation sample. The peak budgets under test were fixed from discovery and not re-selected. We generated 540,000 decodes—270 cap-partitioned shards of 2,000 records each, covering both models, all four arms, three languages, and $k = 8$ —over $B \in \{64, 128, 192, 256, 384, 512, 768, 1024, 2048\}$ together with the premium-scaled caps $\lfloor r_{m,L} B \rfloor$ for NATIVE. The seed varies with the cap, so no two budgets replay one trajectory. We verified all 270 shards; no trace exceeds its cap.

Scoring. We use strict prefix-only exact match on ##### <integer> under intention to treat; truncated, non-integer, and non-compliant answers score 0. Each of the eight samples per item is scored independently at every budget, not by best-of- k , pass@ k , or majority vote. Reported accuracy averages all item-sample cells, and the item-clustered bootstrap resamples the 250 MGSM items while retaining all eight samples within each selected item.

Estimand. Let acc_N and acc_T denote NATIVE and TRANSLATE-ACT accuracy, let $gap(B) = acc_T(B) - acc_N(B)$, and let $r_{m,L}$ be the FLORES-200 token premium of language L over English for

model m . Then

$$\begin{aligned}\Delta_L(B) &= [\text{acc}_T(B) - \text{acc}_N(B)] \\ &\quad - [\text{acc}_T(B) - \text{acc}_N(\lfloor r_{m,L}B \rfloor)] \\ &= \text{acc}_N(\lfloor r_{m,L}B \rfloor) - \text{acc}_N(B). \quad (1)\end{aligned}$$

TRANSLATE-ACT cancels. What remains is simply how many more NATIVE answers become correct between the matched budget B and the premium-scaled budget $\lfloor r_{m,L}B \rfloor$. We call a budget *binding* when correct answer lines are still appearing inside that interval, and the *budget-binding regime* is the range of B over which they are.

This identity has three direct consequences. First, $\Delta_L(B)$ is a finite, discrete increment of the NATIVE accuracy curve, not a comparator interaction. Second, its peak location follows the NATIVE answer-emission distribution, while its height depends on the premium-scaled interval and the native curve inside it. Third, $\Delta_L(B) \rightarrow 0$ once NATIVE accuracy saturates, so a near-zero value above score saturation follows analytically from the estimand.

Primary test. The frozen evaluation budget is $B^* = 1024$, the largest $B \in \{512, 1024\}$ with every $\lfloor r_{m,L}B \rfloor \leq 4096$. The confirmatory family contains six Holm-corrected Qwen tests: H1-existence, directional H1-SESOI ($\Delta_L(B^*) > 5$ points), H2 (a larger effect for Thai than German), and H3 (a dollar-matched strategy reversal) for each of the three languages. Inference uses an item-clustered paired bootstrap (10,000 resamples), a studentized sup- t maximum statistic, and a pre-specified $1.3\times$ tail-conservatism factor; Appendix D gives details. The exploratory two-sided equivalence statement below is separate. The analysis protocol was internally frozen before confirmatory scoring; this was not a public preregistration.

FLORES-200 premiums are Qwen de 1.56 / th 2.55 / sw 1.94 and Llama de 1.58 / th 2.19 / sw 1.93. For each model and language, $r_{m,L}$ is the total-token ratio to parallel English over all 1,012 FLORES-200 devtest sentence pairs after NFC normalization, computed with that model’s own tokenizer and a paired percentile bootstrap over sentence pairs.

4 Regime-dependent results

4.1 The frozen test yields no confirmatory support

At $B^* = 1024$, $\Delta_L(B^*)$ is near zero for every Qwen language, and all six Holm-

family tests fail to reject (formal outcome `no_confirmatory_h1_support`):

- Qwen $\Delta_L(B^*)$: de 0.00, th 0.15, sw 0.05 points. H1-existence raw $p = 0.060$ at Holm-local $\alpha = 0.0083$; H1-SESOI raw $p = 1.0$.
- Llama $\Delta_L(B^*)$: de/th/sw all 0.00. This procedurally matched secondary analysis is outside the confirmatory family and also rejects nothing.

The two non-rejections have different descriptive causes. Qwen NATIVE accuracy plateaus at 79.0% / 47.1% / 33.7% for de/th/sw; between 1024 and the Thai FLORES-scaled budget of 2611, only 3 of 2000 traces become newly correct. Llama NATIVE instead plateaus near floor at 13.6% / 3.9% / 29.0%, alongside never-emission rates of 80.2% / 93.2% / 46.0%. Qwen therefore reaches score saturation at moderate accuracy, whereas Llama de/th often never produces a parseable native answer. Neither pattern implies that all traces have terminated.

4.2 Tight budgets expose a large, then vanishing, artifact

Table 1 places the largest retrospective $\Delta_L(B)$ inside the budget-binding regime (pointwise item-clustered bootstrap 95% CIs).

The Qwen rows of Table 1 contain three different kinds of peak. German gives the clearest two-arm comparison: at $B = 192$ both arms are well away from floor, and premium-scaling raises NATIVE by 34.2 points.

Thai has the largest peak, and also the largest premium: at $B = 256$, $r = 2.55$ gives a premium-scaled budget of 652, so $\Delta_L(256)$ summarizes the gain across a 396-token interval rather than a contrast at equal output widths.

Swahili is weaker evidence. Its peak occurs where TRANSLATE-ACT scores only 0.60%, so it shows a native-prefix rescue rather than a clean two-arm comparison away from floor. The same caution applies to the German crossover at $B = 128$ (§4.3).

Simultaneous max- $|t|$ 95% bands over the sweep keep every Qwen peak away from zero (de@192 [27.8, 40.6], th@256 [32.3, 45.4], sw@128 [10.7, 19.3]), and peak locations are stable in 89.6%, 100.0%, and 87.9% of bootstrap replicates. At B^* the largest language-specific upper bound is 0.32 points for Qwen and 0.00 for Llama, below the exploratory ± 5 -point equivalence margin.

Table 1: Peak length-normalization effects, where $\Delta_L(B) = \text{acc}_N(\lfloor rB \rfloor) - \text{acc}_N(B)$, together with values at the frozen budgets. Intervals are pointwise item-clustered bootstrap 95% CIs. The independent column is the same estimand on the confirmation sample of §3, evaluated at the discovery peak budget rather than at a re-selected argmax.

model/lang	replay peak $\Delta_L(B)$ (pts)	independent $\Delta_L(B)$	peak B	NATIVE acc. at B	TRANSLATE-ACT acc. at B	$\Delta_L(512)$	$\Delta_L(1024)$
Qwen de	34.2 [30.2, 38.3]	34.65	192	16.10	22.60	2.25	0.00
Qwen th	38.9 [34.7, 43.0]	38.60	256	6.20	47.75	8.85	0.15
Qwen sw	15.0 [12.4, 17.7]	13.70	128	8.70	0.60	0.25	0.05
Llama de	8.4 [7.0, 9.9]	8.50	256	3.85	43.10	0.15	0.00
Llama th	2.3 [1.6, 3.1]	2.10	192	1.10	16.90	0.15	0.00
Llama sw	18.2 [15.9, 20.6]	17.65	256	8.25	44.60	1.60	0.00

The peaks survive independent capping. The confirmation sample carries its own frozen family of six Holm-corrected Qwen tests: three one-sided SESOI tests that $\Delta_L(B) > 5$ points at the discovery peak budget, and three two-one-sided equivalence tests that $|\Delta_L(1024)| < 5$. All six reject at family-wise $\alpha = 0.05$ (formal outcome confirmatory_support). The independent peak estimates are 34.65 points for German at $B = 192$, 38.60 for Thai at 256, and 13.70 for Swahili at 128, with standard errors 2.10, 2.26, and 1.37 and $p = 0.0001$ throughout. Each falls inside the published discovery interval. The equivalence estimates at $B^* = 1024$ are 0.15, -0.25 , and -1.25 points, the largest of their three p-values being 0.0003. Peak location replicates as well: the independent argmax sits on the predicted budget in all three Qwen languages. This is a second frozen family rather than a re-run of the first. The $B^* = 1024$ family of §4.1 still fails to reject, and that result is unchanged. What the replication rules out is that the tight-budget magnitudes were an artifact of scoring one stored trajectory at many cut points.

Llama remains outside the confirmatory family and supports no confirmatory claim. Its independent estimates at the discovery peaks are 8.50 points for German, 2.10 for Thai, and 17.65 for Swahili, against discovery values of 8.35, 2.30, and 18.20. The Thai SESOI test does not reject, which the protocol predicted before any scoring: a discovery estimate of 2.30 points cannot clear a 5-point SESOI. Llama German and Thai each shift their argmax by one grid point, and the protocol had likewise flagged those cells as too flat to fix a location—Thai reads 2.20, 2.30, and 2.00 at 128, 192, and 256. A length check supports reading the two frames as one process: independently decoded output-length distributions match the truncated discovery distributions to a median 0.11% of the cap for Qwen and 0.16% for Llama.

Normalizer sensitivity. The behavioral trace-

length ratio is smaller than the FLORES premium in all six model-language cells, with non-overlapping pointwise intervals, so FLORES consistently grants the larger premium (Appendix C).

Substituting the behavioral ratio would still produce a 5-point artifact in four of the six cells. The minimum evaluated premium that produces a 5-point artifact lies below the behavioral ratio for Qwen de (1.089 vs 1.467) and th (1.188 vs 2.038), and for Llama de (1.274 vs 1.432) and sw (1.253 vs 1.848). Qwen Swahili is the exception: its threshold of 1.254 sits above the behavioral ratio of 1.179, so the 5-point Swahili claim depends on a premium above that ratio, including the frozen FLORES token premium. Llama Thai never reaches 5 points, even at $1.5 \times$ FLORES.

These thresholds are grid-selected and conditional on the stored traces. The behavioral ratio is not a validated normalizer, because it compares traces that differ in content, correctness, and stopping behavior.

Figure 1 plots the Qwen NATIVE curves and marks both accuracy endpoints of each peak premium interval.

4.3 Tight caps can reverse strategy rankings

Table 2 shows that Qwen crossover strength follows the left tail of answer-emission timing more closely than the medians.

TRANSLATE-ACT’s median emission is earlier than NATIVE’s in German and Thai. The p10 ordering is timing-consistent with the crossover. Swahili has a 74.7-token NATIVE early-tail advantage and the strongest native lead. German differs by only 2 tokens and has a narrow lead, and Thai NATIVE is 71 tokens later and never leads. These grid-resolved emission summaries do not isolate translation-segment length or establish mediation.

At $B = 128$, Qwen German NATIVE scores 2.55% versus 1.15% for TRANSLATE-ACT; Swahili NATIVE leads with bootstrap probability

Qwen3-8B native accuracy and peak premium windows

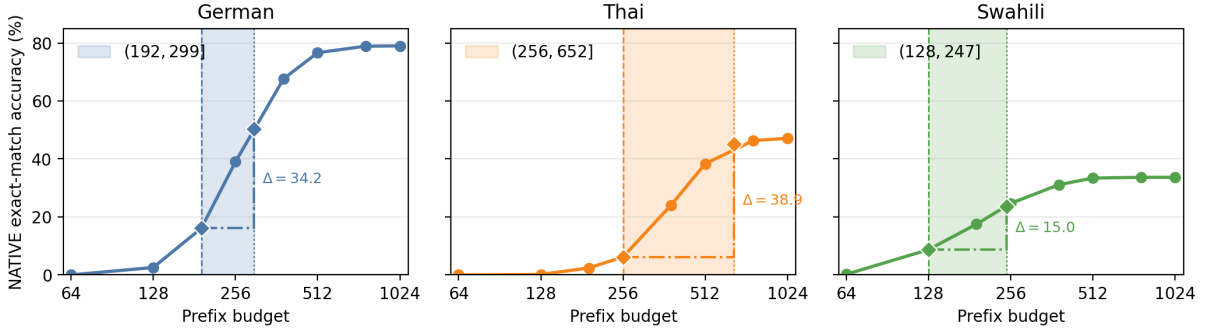


Figure 1: Qwen NATIVE accuracy by budget. Shaded intervals mark each language’s peak normalization interval ($B_{\text{peak}}, \lfloor rB_{\text{peak}} \rfloor$]; the vertical gain between the endpoint markers equals the peak $\Delta_L(B)$.

Table 2: Qwen answer-emission timing and observed strategy crossovers. E is the token position at which a parseable answer is emitted; p10 is its 10th percentile.

lang	NATIVE median E	TRANSLATE-ACT median E	NATIVE p10 E	TRANSLATE-ACT p10 E	observed crossover
sw	206	262	96	170.7	strongest: NATIVE leads at 64/128/192
de	270	247	160	158	marginal: NATIVE leads at 128
th	377	250	236	165	none

1.00 at 128 and 192 (transition [192,256]), while German leads with probability 0.958 at 128 (transition [128,192]). Thai has no native lead, and no Llama language crosses because native accuracy remains near floor. The Qwen Swahili crossover also comes from a degenerate heavy-tail cell: NATIVE output-length p90 is 4096 and 25.1% never emit a parseable answer. The frozen H3 test did not detect these reversals because it examined only the looser pre-specified budgets.

5 Measurement audits

We report five ledger audits and one diagnostic. Trace-language ID, COMET, parser robustness, and decoder parity appear below; normalizer sensitivity appears in §4.2. The verbosity/failure-tail decomposition is a diagnostic, not a sixth audit.

Trace-language ID (automated GlotLID). Automated labels support the main NATIVE vs TRANSLATE-ACT contrast. Determinate NATIVE traces are classified in L at Qwen de 92.1% / sw 94.1% / th 99.4%, and at approximately 100% for all Llama languages; TRANSLATE-ACT post-delimiter reasoning is 98.4–99.9% English.

Three caveats apply. First, the initial sw-only mapping failed the frozen native:sw agreement criterion ($75\% < 90\%$). We then adopted the Swahili macrolanguage remap (sw+swc, with swc denoting Congo Swahili), which raised Qwen native-Swahili compliance from 85.8% to 94.1% and the

validation cell to 90.00% (18/20). Because we adopted that remap after inspecting the failure, we report it as a post-hoc analytic decision. Unrelated neighboring Bantu labels remain outside Swahili. Second, an independent blind LLM adjudication of 240 Qwen traces agrees at 96.7% overall and at least 90% in every cell, but it is preliminary and the frozen human validation remains outstanding. Third, PIVOT and CODE-SWITCHED violate their English instruction in 9 of 12 cells, so they remain outside the main comparison.

COMET translation quality. Reference-based COMET is generally high but nonuniform (Table 4): German is strongest for both models (0.877 Qwen, 0.872 Llama), Qwen Swahili is lowest at 0.749, and Llama Thai has a weak lower tail (p10 0.325). At the six prespecified peak budgets, an exploratory per-cell analysis finds weak and inconsistent COMET–correctness-gain associations (Spearman $\rho = -0.143$ to 0.280, five positive, only one with $|\rho| \geq 0.20$; Table 5), so the tight-budget TRANSLATE-ACT advantage is not well explained by translation quality. Translation quality remains part of the strategy bundle; these scores are descriptive and never condition accuracy.

Parser robustness. In NATIVE peak cells, prefix-only rescued-correct traces are at most 0.35% and value-unstable traces at most 0.30%. Within the $(B, \lfloor rB \rfloor]$ intervals, 96.8–100% of native gains are genuinely terminated, and requiring

a terminated answer line changes every peak by at most 0.2 points. The tight-budget effect is therefore late answer emission rather than a prefix-parser artifact.

Decoder parity. A stratified 2,520-prefix audit finds 37.9% raw exact-string agreement between local and vLLM decoding but 100% agreement after production special-token normalization, including 100% parsed-answer and correctness agreement. All raw divergences are cosmetic special-token markup.

Verbosity/failure-tail diagnostic. Qwen Swahili NATIVE has a heavy tail: 10.6% hit the 4096 cap and 25.1% never emit a parseable answer. This mixes truncation, non-integer or multiple answers, and format noncompliance, cautioning against interpreting native accuracy as pure reasoning ability.

6 Implications for adaptation

A natural response to a measured multilingual gap is a cost-ordered ladder of fixes: raise the serving budget, then change the prompting strategy, then add language-specific tokens to the tokenizer, and finetune only if all three fail. Our ledger prices the two token-count rungs.

Both token-count rungs act only by relieving truncation. A larger cap and a cheaper tokenizer both change one thing: how much of the trace the model is allowed to finish. Where the trace already fits, neither can change an answer. This is the same channel Equation (1) describes. Writing ρ for the factor by which an extension shortens target-language text, a NATIVE-only payoff of $\text{acc}_N(\lfloor \rho B \rfloor) - \text{acc}_N(B)$ has exactly the form of Equation (1) with ρ in place of $r_{m,L}$. We use that only as motivation, not as a result: compression is not uniform across traces, and a deployed extension changes both arms, so we measure the effect directly rather than substituting ρ into the identity.

Measurement, not extrapolation. We built genuinely extended Qwen tokenizers—base vocabulary and merge list untouched, new byte-level merges learned on target-language traces and appended so that base merges retain priority—and then measured what they buy under a cap. Each stored trace is retokenized with the extended tokenizer, its first B extended token ids are decoded, and that text is scored with the same strict parser used throughout the paper. The baseline is the identical computation with the base tokenizer, and

no uniform compression factor is applied. Training text is the NATIVE arm only: the PIVOT and CODE-SWITCHED arms are substantially English in several cells (Swahili PIVOT is 90.1% English, §5) and would not train a language-specific vocabulary. Extensions are cross-fitted over two disjoint halves of the item set, so no evaluated item contributed a merge, and the extension size is fixed in advance by a rule that never consults accuracy—the largest extension each fold’s corpus admits. This yields roughly 3.4k, 6.7k and 3.1k new tokens for de/th/sw, which cut the FLORES-200 devtest premium from 1.559 to 1.531, from 2.551 to 2.207, and from 1.936 to 1.846. The aggregate English devtest token count changes by less than 0.02% (Appendix F).

The extension is applied to both arms, so G_3 reflects one deployment change rather than a NATIVE-only advantage. Because TRANSLATE-ACT traces are largely English, a target-language extension barely compresses them; their gains are correspondingly small, which is a property of the intervention rather than an artifact of the design.

This is a token-count-only counterfactual, not a prediction about a retrained model. It holds the emitted text fixed and changes only what that text costs. A real extension must also learn embeddings for the new tokens (Cui et al., 2023; Fujii et al., 2024; Dobler and de Melo, 2023), which is itself a training procedure, and a model whose tokenizer changed would follow a different trajectory.

Table 3 prices the ladder, and three patterns cut against the monotone $G > G_1 > G_2 > G_3$ expectation.

First, *more budget need not shrink the gap*. At $B = 128$ the measured deficit is negative in German and Swahili and near zero in Thai, because TRANSLATE-ACT has not finished its translation preamble (§4.3); at the largest stored prefix Swahili moves from $G = -8.10$ to $+23.30$. These cells sit near the accuracy floor—both arms are below 10% at $B = 128$ —so this is an exploratory sign instability rather than a calibrated estimate of harm. It is nonetheless enough to show that a gap measured under a tight cap can carry the wrong sign.

Second, *vocabulary extension captures only a fraction of what a larger cap would*. The informative comparison is not the truncated share but $\text{acc}_N(4096) - \text{acc}_N(B)$, the NATIVE accuracy a longer prefix actually recovers: truncated traces that were never going to be scored correct cannot be rescued by any token-count intervention. Against

Table 3: Adaptation triage for Qwen. G is the NATIVE deficit against TRANSLATE-ACT at the deployed cap B ; $G(4096)$ is the deficit at the largest stored prefix; G_3 is the deficit after a cross-fitted language-specific vocabulary extension applied to both arms at the original cap. “NAT. trunc.” is the share of NATIVE traces longer than B ; “NAT. gain” is $\text{acc}_N(4096) - \text{acc}_N(B)$, the NATIVE accuracy actually recovered by extending the prefix to the largest one stored. It is observed on this ledger, not a general ceiling: 4096 still binds, and it does not bound gap closure, which also depends on TRANSLATE-ACT. Gap closure carries item-clustered bootstrap 95% CIs, conditional on the two fitted tokenizers. The prompting rung is not priced: the comparator TRANSLATE-ACT *is* that rung, so it closes G by construction.

lang	B	NAT. trunc.	NAT. gain	G	$G(4096)$	G_3 (vocab)	gap closed by vocab [95% CI]
de	128	97.1%	76.45	−1.40	+9.40	−2.30	+0.90 [−0.05, 1.85]
de	256	52.1%	40.00	+10.25	+9.40	+9.05	+1.20 [−0.25, 2.65]
de	512	3.9%	2.35	+10.95	+9.40	+10.45	+0.50 [0.05, 1.10]
de	1024	0.2%	0.00	+9.35	+9.40	+9.35	+0.00 [0.00, 0.00]
th	128	99.9%	47.10	+0.75	+40.60	−0.10	+0.85 [0.20, 1.65]
th	256	84.7%	41.05	+41.55	+40.60	+38.15	+3.40 [1.90, 5.00]
th	512	18.9%	8.90	+48.40	+40.60	+43.50	+4.90 [3.65, 6.20]
th	1024	0.7%	0.15	+40.75	+40.60	+40.75	+0.00 [0.00, 0.00]
sw	128	81.5%	25.05	−8.10	+23.30	−9.40	+1.30 [0.75, 1.90]
sw	256	42.6%	9.00	+5.00	+23.30	+4.35	+0.65 [−0.20, 1.55]
sw	512	15.3%	0.35	+22.75	+23.30	+22.80	−0.05 [−0.25, 0.15]
sw	1024	11.6%	0.10	+23.40	+23.30	+23.35	+0.05 [0.00, 0.15]

that quantity, the extension’s own NATIVE gain is 3.45 of 40.00 points for German at $B = 256$ (9%) and 5.20 of 41.05 for Thai (13%), rising to 5.05 of 8.90 for Thai at $B = 512$ (57%). Its largest gap closure anywhere in our data is 4.90 points, and its largest NATIVE gain is 5.20. Where the longer prefix recovers nothing the extension does too: at $B = 1024$ the NATIVE gain to 4096 is 0.00 / 0.15 / 0.10 points and closure is 0.00/0.00/0.05. German at $B = 256$ and Swahili at $B = 256$ and 512 have intervals that include zero.

The Swahili rows show why the truncated share alone is misleading. At $B = 1024$, 11.6% of NATIVE traces exceed the cap, but 211 of those 232 traces run to the 4096-token generation cap without stopping, only 23 are parseable even from the full stored trace, and only two become correct by 4096. The NATIVE gain to 4096 is therefore 0.10 points, of which the extension recovers 0.05. The cell is not evidence that extension is weak against real truncation; it is a cell with almost nothing left to recover within the ledger, and reporting truncation without the realized gain would have obscured that.

Third, *the residual gap is the whole gap*. At the largest stored prefix, 9.4 / 40.6 / 23.3 points remain for de/th/sw. We cannot report the prompting rung as a validated intervention, because adopting TRANSLATE-ACT closes G by definition; its standalone value would have to be established against a different comparator.

Two cautions constrain how far this general-

izes. Our ledger caps generation at 4096 tokens, so $G(4096)$ is the gap at the largest prefix we stored, not a demonstrated non-binding regime; 10.55% of Swahili NATIVE generations reach that cap without stopping. The scale of the intervention is also modest by construction. Cutting the Thai premium from 2.551 to 2.207 is 13.5% fewer tokens for the same text, equivalently 15.6% more text under a fixed cap, against the four- to thirty-two-fold increases available by raising the cap itself.

The practical guidance is therefore narrow. Before paying for either token-count rung, measure the realized gain directly: extend the cap on a sample and see how much accuracy the longer prefixes actually recover. Where that gain is small, neither rung is likely to pay, however many traces the cap truncates—though this is dispositive only if the extended cap is itself non-binding, which ours is not. Where the gain is large, our data do not support a general ordering. Raising the cap dominates for *native accuracy*—40.00 against 3.45 points for German at $B = 256$ —but not necessarily for the *gap*, because a larger cap lifts TRANSLATE-ACT too. German gap closure is 1.20 points from extension against 0.85 from raising the cap, and Swahili’s gap widens from 5.00 to 23.30. Vocabulary extension may also reduce decoding steps, though we did not measure net serving cost or latency, and a larger embedding and output projection work against the saving. We did not test the finetuning rung, the prompting rung is confounded with our compara-

tor, and this is one model family on one benchmark, so this is a triage heuristic derived from our estimand, not a validated adaptation method.

7 Scope and implications

In these stored-prefix MGSM evaluations, multilingual exact-match comparisons are regime-dependent: length normalization has large descriptive effects while answer emission binds, and negligible effects after score saturation. The prospectively frozen non-rejection at $B^* = 1024$ is a negative boundary condition. The retrospective sweep locates where the sensitivity actually occurs, and its six pre-registered cells then held on independently capped decodes (§4.2). The remainder of the sweep is still exploratory.

At the larger evaluated budgets the residual native-vs-translate difference is large (Qwen Thai about +41 points, Llama Thai about +69), but it is a strategy-performance gap. Prompt language, reformulation, format compliance, translation quality, and reasoning language remain confounded.

Calling this residual “not an identified reasoning deficit” states what our design can identify, not that execution effects are absent: complementary diagnostics find genuine target-language reasoning-execution failures even with English inputs (Zhang et al., 2026).

Limitations. (i) The tight-budget magnitudes no longer rest on prefixes of one long decode: the three Qwen peak cells and the equivalence at B^* were pre-registered and confirmed on independently capped generations (§4.2). The rest of the sweep is still retrospective and exploratory—the crossovers in §4.3, the grid points outside those six cells, and the normalizer-sensitivity analysis. That replication removes the shared-trajectory artifact without testing budget-aware generation. Under our serving stack `max_tokens` only stops decoding and never conditions the model: with a shared seed, 75% of capped decodes come back bitwise identical to the truncated long decode. Neither frame therefore speaks to a deployment that announces the budget in the prompt or forces an answer when the cap arrives. We have not tested whether a model told its budget in advance would behave differently. (ii) vLLM repeat generation was only 46% bitwise deterministic, preserving the stored-ledger estimand but weakening reproducibility and shared-seed interpretation. (iii) Scope is MGSM, three languages, and two 8B models. (iv)

The frozen human GlotLID validation remains outstanding, despite the preliminary blind LLM agreement. (v) The frozen same-content trace-premium validation was not performed; the behavioral ratio in Appendix C does not substitute for it. (vi) Calibration is only approximately nominal (corrected type-I 0.00917 vs 0.00833 target); the $1.3\times$ factor is a family-wide safeguard, not a verified family-wise calibration. (vii) The vocabulary-extension results in §6 are a token-count-only counterfactual that does not predict a retrained model, whose trajectory would differ and whose new embeddings must themselves be learned. They cover Qwen alone, because the extension is tokenizer-specific. The extensions are trained on MGSM reasoning traces rather than general text, so the compression we observe is a property of this corpus and not a bound on what is achievable. Appending merges preserves base-merge priority but not every English segmentation, and we did not measure net serving cost or latency. The bootstrap resamples items while holding the two cross-fitted tokenizers fixed, so the intervals are conditional on that draw and do not propagate the uncertainty of learning the merges. The fold split is fixed rather than randomized, and we did not run repeated splits. Both arms in Table 3 are scored by retokenizing the stored text rather than by replaying stored token ids. The two paths agree on 59,998 of 60,000 sample-budget comparisons and differ by at most 0.10 accuracy points in any reported cell. $G(4096)$ is the gap at the largest stored prefix, not a demonstrated non-binding regime. The emission percentiles quoted in §6 are computed among emitting traces only, covering neither the 1.95–25.1% that never emit nor the upper tenth that emit late. The $B = 128$ sign reversals sit near the accuracy floor.

Future work should test prompt interventions that elicit earlier answer emission—which would shrink NATIVE’s budget-binding regime—and extend the protocol beyond MGSM numeric exact match to longer-form and non-numeric tasks.

Data and code availability. The full ledger of stored generations and all analysis code will be released.

Takeaway. When comparing language strategies under a token cap, treat the cap as an independent variable rather than a background constant. Report accuracy across a budget sweep with a length-normalized comparison, mark the region in which answer emission binds, and do not let a single budget carry the claim.

References

- Orevaoghene Ahia, Sachin Kumar, Hila Gonen, Jungo Kasai, David R. Mortensen, Noah A. Smith, and Yulia Tsvetkov. 2023. Do all languages cost the same? tokenization in the era of commercial language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9904–9923.
- Yiming Cui, Ziqing Yang, and Xin Yao. 2023. [Efficient and effective text encoding for chinese llama and alpaca](#). *arXiv preprint arXiv:2304.08177*.
- Konstantin Dobler and Gerard de Melo. 2023. [FOCUS: Effective embedding initialization for monolingual specialization of multilingual models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 13440–13454.
- Kazuki Fujii, Taishi Nakamura, Mengsay Loem, Hiroki Iida, Masanari Ohi, Kakeru Hattori, Hirai Shota, Sakae Mizuki, Rio Yokota, and Naoaki Okazaki. 2024. [Continual pre-training for cross-lingual LLM adaptation: Enhancing japanese language capabilities](#). *arXiv preprint arXiv:2404.17790*.
- Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc’ Aurelio Ranzato, Francisco Guzmán, and Angela Fan. 2022. The flores-101 evaluation benchmark for low-resource and multilingual machine translation. *Transactions of the Association for Computational Linguistics (TACL)*, 10:522–538.
- Aaron Grattafiori et al. 2024. [The llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.
- Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schütze. 2023. GlotLID: Language identification for low-resource languages. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6155–6218.
- Hyunwoo Ko, Guijin Son, and Dasol Choi. 2025. [Understand, solve and translate: Bridging the multilingual mathematical reasoning gap](#). *arXiv preprint arXiv:2501.02448*.
- Maxence Lasbordes, Amélie Chatelain, and Djamé Seddah. 2026. [Rethinking the multilingual reasoning gap with layer swap](#). *arXiv preprint arXiv:2605.26735*.
- Yihong Liu, Raoyuan Zhao, Michael A. Hedderich, and Hinrich Schütze. 2026. [Crosslingual on-policy self-distillation for multilingual reasoning](#). *arXiv preprint arXiv:2605.09548*.
- Wenyang Luo, Wayne Xin Zhao, Jing Sha, Shijin Wang, and Ji-Rong Wen. 2025. [MMATH: A multilingual benchmark for mathematical reasoning](#). *arXiv preprint arXiv:2505.19126*.
- Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. 2025. [s1: Simple test-time scaling](#). *arXiv preprint arXiv:2501.19393*.
- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Hefernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, et al. 2022. [No language left behind: Scaling human-centered machine translation](#). *arXiv preprint arXiv:2207.04672*. Introduces the FLORES-200 benchmark.
- Aleksandar Petrov, Emanuele La Malfa, Philip Torr, and Adel Bibi. 2023. Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.
- Qwen Team. 2025. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.
- Ricardo Rei, José G.C. de Souza, Duarte Alves, Chrysoula Zerva, Ana C. Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and André F.T. Martins. 2022. COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585.
- Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2023. [Language models are multilingual chain-of-thought reasoners](#). In *The Eleventh International Conference on Learning Representations (ICLR)*.
- Gianluigi Silvestri and Edoardo Cetin. 2026. [Learning from partial chain-of-thought via truncated-reasoning self-distillation](#). *arXiv preprint arXiv:2603.13274*.
- Ian Su, Gaurav Purushothaman, Jey Narayan, Ruhika Goel, Kevin Zhu, Sunishchal Dev, Yash More, and Maheep Chaudhary. 2026. [Broken chains: The cost of incomplete reasoning in LLMs](#). *arXiv preprint arXiv:2602.14444*.
- Jiaqiao Zhang, Zhoujun Li, Raoyuan Zhao, Jian Lan, Thomas Seidl, Michael A. Hedderich, Hinrich Schütze, and Yihong Liu. 2026. [Beyond input understanding: Diagnosing multilingual mathematical reasoning with directed acyclic trace graphs](#). *arXiv preprint arXiv:2605.27715*.

A Accuracy-curve summary

The complete curves use $B \in \{64, 128, 192, 256, 384, 512, 768, 1024\}$. Qwen NATIVE accuracy rises from 0.0/0.0/0.2% at $B = 64$ to 79.0/47.1/33.7% at $B = 1024$ for de/th/sw; Llama NATIVE rises from 0.0/0.0/0.0% to 13.6/3.9/29.0%. The Qwen curves climb sharply through the peak intervals in Figure 1 and then flatten, whereas the Llama de/th curves remain

near floor. The supplementary material reports every model–language–arm value at all eight budgets.

B Best-English-arm

Reselecting the empirically best English arm inside each bootstrap replicate reproduces the preselected TRANSLATE-ACT gap closely (Qwen Thai about 41 points, Llama Thai about 69; uplift over TRANSLATE-ACT is near zero in most cells).

C Trace-length ratio (not a normalizer)

The ratio of median NATIVE output tokens to median TRANSLATE-ACT post-delimiter English-reasoning tokens is Qwen de 1.47 / th 2.04 / sw 1.18. Subtracting the FLORES premium gives a negative difference in all six cells, with non-overlapping pointwise intervals: Qwen de $[-0.092, -0.144]$, th $[-0.513, -0.588]$, sw $[-0.757, -0.824]$; Llama de $[-0.149, -0.193]$, th $[-0.427, -0.488]$, sw $[-0.083, -0.129]$. This ratio compares behaviorally different traces and cannot validate or replace FLORES.

D Statistical machinery and the six-test family

The Qwen confirmatory family contains exactly six raw p-values, corrected by Holm at family level $\alpha = 0.05$: H1-existence tests whether any language has $\Delta_L > 0$; H1-SESOI separately tests whether any has $\Delta_L > 5$ points; H2 tests the directional ordered contrast $\Delta_{th} - \Delta_{de} > 0$; and H3-de, H3-th, and H3-sw each test for a dollar-matched strategy reversal, requiring a significantly positive NATIVE-minus-TRANSLATE-ACT contrast at one common-support budget and a significantly negative contrast at another. Each H3 p-value is the maximum of the multiplicity-controlled positive- and negative-side p-values, an intersection-union test.

Inference resamples the 250 parallel MGSM items as clusters, retaining all languages, arms, and eight samples within each selected item, for 10,000 paired bootstrap resamples. Studentized sup- t maxima provide simultaneous bounds over languages for H1 and over budgets for H3; H2 uses the corresponding one-sided studentized bootstrap contrast. Because calibration with 250 discrete item clusters found mild anti-conservatism in the extreme $\alpha/6$ tail, every raw confirmatory p-value is multiplied

by the pre-specified 1.3 tail-conservatism factor (capped at 1), equivalently testing at $\alpha/1.3$, before Holm correction.

The independent-decoding replication has its own protocol and its own family of six tests, specified in prereg-independent-decoding.md at tag independent-protocol-freeze, with results in analysis-out/independent_scoring.json. It reuses this machinery unchanged, except that $acc_N(B)$ and $acc_N(\lfloor rB \rfloor)$ now come from different generations and so are paired within item rather than within trace.

E TRANSLATE-ACT translation quality (COMET, descriptive)

Reference-based Unbabel/wmt22-comet-da; one stored full trace per item (sample 0); delimiter-missing traces excluded from this denominator only; pointwise 95% item-bootstrap CIs (10,000 resamples).

These scores are descriptive only and never condition, gate, or reweight accuracy.

F Vocabulary-extension measurement

The extension procedure is described in §6. Cross-fitting runs over two disjoint halves of the 250 MGSM items, so no evaluated item contributed a merge to the tokenizer that scores it, for either arm. The base pipeline reproduces the frozen premiums exactly (1.5589 / 2.5508 / 1.9363).

The English control compares aggregate English devtest token counts before and after extension. It bounds the aggregate cost below 0.02%, but appending merges guarantees only that base merges retain priority; it does not guarantee that every English string segments identically, and a small number do change.

Extension sizes are limited by our corpora, which are MGSM reasoning traces rather than general target-language text; a production-scale extension trained on a broader corpus would plausibly compress more, but we cannot bound that from these data, so we report the compression observed under this corpus rather than a lower bound on what is achievable.

Accuracy in Table 3 does not use ρ as a multiplier; the retokenization path is the one described in §6. Decoding token ids, rather than slicing the source string at a token offset, matters because byte-level tokens can split a code point, so an offset slice would occasionally admit slightly more than B to-

Table 4: Reference-based COMET translation quality for TRANSLATE-ACT. Intervals are pointwise 95% item-bootstrap CIs.

Model	Lang	n	Missing delimiter	COMET mean	median	p10	p90	95% CI
Qwen	de	250	0.0%	0.877	0.878	0.834	0.921	[0.872, 0.881]
Qwen	th	249	0.4%	0.858	0.866	0.810	0.907	[0.851, 0.865]
Qwen	sw	250	0.0%	0.749	0.772	0.580	0.867	[0.734, 0.763]
Llama	de	249	0.4%	0.872	0.874	0.826	0.915	[0.868, 0.877]
Llama	th	244	2.4%	0.783	0.850	0.325	0.895	[0.759, 0.805]
Llama	sw	240	4.0%	0.798	0.825	0.703	0.888	[0.783, 0.811]

Table 5: Exploratory COMET association with correctness gain at each prespecified peak token budget, using sample 0. Intervals are pointwise bootstrap 95% CIs.

Model	Lang	Peak B	n	Spearman ρ [95% CI]
Qwen	de	192	250	0.015 [-0.112, 0.139]
Qwen	th	256	249	0.146 [0.019, 0.269]
Qwen	sw	128	250	-0.143 [-0.269, -0.004]
Llama	de	256	249	0.280 [0.155, 0.399]
Llama	th	192	244	0.009 [-0.129, 0.146]
Llama	sw	256	240	0.074 [-0.051, 0.203]

Table 6: Qwen vocabulary extension, per cross-fitting fold. r is the frozen FLORES-200 premium and r' is the premium under the extended tokenizer. The English control is the aggregate English devtest token count relative to the base tokenizer. These are the extensions used in Table 3.

Lang	Fold	New tokens	r	r'	English
de	0	3,470	1.559	1.531	0.99996
de	1	3,343	1.559	1.532	0.99996
th	0	6,798	2.551	2.195	1.00000
th	1	6,577	2.551	2.218	1.00000
sw	0	3,173	1.936	1.865	0.99993
sw	1	3,054	1.936	1.826	0.99986

kens. Scoring generated text rather than stored token ids is a reimplementa-
tion of the scorer used
elsewhere in the paper, so we verified it against that
scorer on all six Qwen cells at five budgets. The
two paths agree on 59,998 of 60,000 sample-budget
comparisons, the two exceptions being Swahili NA-
TIVE traces whose text does not round-trip to its
stored token ids. Every column of Table 3 is pro-
duced by the retokenization path alone, so the re-
ported closure is exactly $G - G_3$. No model weights
are trained.